Bojja Divya

+91 6302877035 | divyabojja9933@gmail.com | linkedin.com/in/bojja-divya | github.com/bdivya-9125

EDUCATION

Vellore Institute of Technology *Chennai, India*
Bachelor of Technology in Electronics and Communication Engineering 2023 – 2027

- CGPA: 9.08 / 10

EXPERIENCE

PCB Design & Hardware Validation Intern | Suryas MiB Enterprise *Aug 2025 – Sep 2025*

- Designed and simulated circuit board layouts using NI Multisim, covering schematic capture, component modeling, and signal-level validation and test for system prototypes.
- Analyzed BJT, MOSFET, LED, and 555-timer circuits under varied bias and load conditions to characterize circuit behavior, power consumption, and signal integrity.
- Performed structured hardware debugging on simulated prototypes and authored engineering test documentation covering thermal, noise, and EMI considerations relevant to board-level design.

PROJECTS

Real-Time Glucose Monitoring and Embedded Medical Advisory System
Arduino, 8051 Microcontroller, UART, Embedded C, I2C LCD, Hardware Bring-up

- Built an end-to-end embedded hardware system integrating a potentiometer sensor, Arduino (ADC), and an 8051 microcontroller for real-time signal acquisition, processing, and classification.
- Assembled and brought up the hardware prototype on breadboard (Arduino, 8051 development board, I2C LCD, potentiometer), performing component-level wiring, functional verification, and hardware debugging.
- Implemented UART-based inter-microcontroller communication between Arduino and 8051, and a rolling-average filtering algorithm to reduce signal noise and improve measurement stability.
- Interfaced the embedded system with a Python-based web dashboard for live telemetry, extending the hardware pipeline to remote monitoring.

Comparative Study of 16-bit Adder Architectures
Cadence Virtuoso, VLSI, GPDK 180nm

- Designed and simulated 16-bit Ripple Carry and Weinberger Adders in Cadence Virtuoso, applying industry-standard circuit design and validation methodology at the silicon level.
- Ran transient simulation and analyzed propagation delay, power consumption, and architecture efficiency — core signal integrity and power design trade-offs.
- Achieved 10–20x lower delay with the Weinberger Adder versus the Ripple Carry architecture, quantifying the performance/power design trade-off.

Hand Gesture Recognition Glove
Arduino, 8051, Embedded Systems, Bluetooth

- Integrated flex sensors with Arduino and an 8051 microcontroller to capture, condition, and process real-time analog sensor inputs — end-to-end board-to-firmware hardware automation.
- Implemented Bluetooth communication over UART for reliable wireless data transmission and system-level integration testing.
- Developed threshold-based embedded decision logic and validated end-to-end system operation against real hardware, following a structured test approach.

PATENTS

Optimized Standard Cell Layout Generation for VLSI | Indian Patent Office *Published Mar 2026*

- App No: 202641021183 — Proposed an algorithmic framework for automated standard cell layout optimization with reliability-aware modeling of aging, electromigration, and parasitic effects — directly relevant to board- and chip-level design automation and hardware reliability.

System and Method for Cognitive Navigation and Location Verification | Indian Patent Office *Published Jul 2026*

- App No: 202641087731 — Designed an AI-based navigation framework with predictive location modeling to forecast and autonomously verify GNSS positioning reliability and accuracy.

TECHNICAL SKILLS

- **Hardware & PCB Design:** PCB Layout Design, Schematic Capture, System/Board Design, Signal Integrity, Power Design, Circuit Testing, Hardware Debugging, Prototype Assembly & Bring-up, Validation & Test
- **Electronics & Embedded:** Digital & Analog Electronics, Sensors, Embedded Systems, Microcontrollers (8051, Arduino), Wireless Communication, UART, CAN
- **VLSI & Design:** Verilog HDL, RTL Design, VLSI Design, Static Timing Analysis
- **Tools & Software:** Cadence Virtuoso, NI Multisim, MATLAB, Arduino IDE, Keil, ProgISP, Git, GitHub
- **Programming Languages:** C, C++, Python